

Kubernetes

Page No.:

Date 22/3/25

- * Kubernetes is a orchestration tool for managing containers.
- * Features orchestration tools offer.
 - High availability or no downtime
 - Scalability or high performance
 - Disaster recovery - backup & restore

- * I can talk to kube server through 3 ways
 - UI → dashboard
 - API →
 - CLI → kube ~~CTL~~ CTL

Pod

- Smallest unit in Kubernetes.
- Abstraction over container (extra layer over container)
- Usually 1 Application for Pod.
- Each Pod gets its own IP address.
- Pods are ephemeral

Service

- Permanent IP address
- Lifecycle of Pod and Service not connected.

Ingress

- ~~forward~~ converts domain name to its ip address

Configmap

- external configuration of your application (like api)

Secret → external config file which changes
(username, password, security certificates, etc)
Secret file

Volumes

Kubernetes cluster → Storage.

It does not provide data persistent

Deployment → stateless app² (frontend)
Blueprint for 'my-app' Pods (yaml format)

You create Deployments

Abstraction of Pods

→ ~~yaml~~ It will contain all necessary files for deployment.

it contains (configuration, meta-data, status)

→ Service ~~does~~ acts as load-balancing.

Stateful set

- making stateful set is the blueprint for creating pods of stateful application.
- stateful set file is far more difficult than is creating than stateless file (yaml) (frontend)
- STATEFULSET → MySQL, MongoDB,

→ choco install minikube

→ minikube version

→ minikube start --vm-driver=docker → download image 1st time

* if we have run ^{above} this command and it fails then its file become dangling file then it starts itself so delete before re-start minikube.

→ minikube delete --all → to del all dangling files.

→ minikube status → check status of minikube.

→ ~~minikube~~ kubectl get node → all nodes

* minikube is generally used for starting the cluster, getting status of cluster, delete the cluster, stopping the cluster.

→ ~~minikube~~ minikube start

→ minikube ~~delete~~ stop

* kubectl is the CLI interface for creating, managing, running pods and services.

→ kubectl get node → all nodes

* To create Pods we have to create: deployment file, service ^{file}, configmap file, secret file

* 1st demo

running nginx applⁿ on k8s cluster.

for this demo we will create 2 files.

→ deployment & it is blueprint of pods

→ service → used for networking, service

deployment (nginx) it contain 3 files for deployment metadata, specs, template

it has 3 layers

* 1st label → deployment file connected to level of pods

* replicas: contains no's (1, 2, 3, ... 8)

• Template :- all data inside template is pods data (specs of container)

* Service file is used for communication/networking

→ service is of 3 types

• external, internal, load balancer

ports → (NodePort) (ClusterIP) (Load Balancer)
By default

nodePort → (30,000 - 32,767) any value b/w it

Deployment → minikube → Pods.

• Kubectl apply -f nginx.yaml

• Kubectl get all

• minikube service list

• minikube service nginx-service --url

• ~~minikube~~ kubectl get node

• " " " pods

• " " " service

Satyam Kumar

2211981354

A)

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deployment

label:

app: nginx-app

spec:

replicas: 2

selector:

matchLabels:

app: nginx-app

template:

metadata:

labels:

app: nginx-app

spec:

container:

name: nginx

image:

ports:

Container Port: 80

apiVersion: apps/v1

kind: Service

metadata:

name: nginx-service

selector:

app: nginx-app

spec:

type: NodePort

Intro to AWS Kinesis & Lambda Service

Page No.:

Date: 27/7/25

selector:

app: nginx-app

ports:

protocol: TCP

port:

targetPort: 80

nodePort: 8080

B

- minikube version
- minikube start --driver=Docker
- minikube status
- kubectl apply -f nginxapp.yaml
- kubectl get all
- minikube service nginx-service --url
- minikube stop
- minikube delete -all